

Q1. Table to Visual: Mapping data onto aesthetics [10 marks]

(a) Data type of each variable [2]

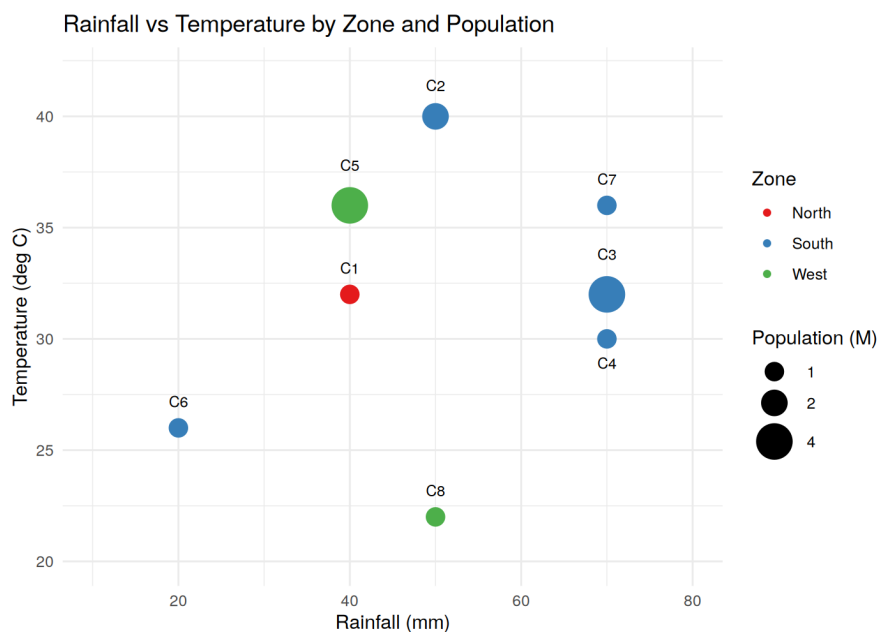
Variable	Data type	Reason
Rainfall_mm	Continuous	Measured numeric quantity (any value in a range).
Temp_C	Continuous	Measured numeric quantity (interval scale).
Zone	Nominal	Names (North, South, West) with no order.
Population_M	Discrete	Recorded only as whole numbers (1, 2, 4) in this table.

(b) Aesthetic and scale type for each variable [3]

Variable	Aesthetic	Scale type (ggplot2)
Rainfall_mm	x position	Continuous - scale_x_continuous()
Temp_C	y position	Continuous - scale_y_continuous()
Zone	colour	Discrete (qualitative) - e.g. scale_colour_brewer(palette = "Set1")
Population_M	size (area of the point)	Continuous size scale - scale_size_area()

Chart chosen: a **scatter plot (bubble chart)** with one point for each city.

(c) Sketch of the chart with labelled axes and legends [3]



Sketch drawn with ggplot2 (the R program is given below). x-axis: Rainfall (mm); y-axis: Temperature (deg C). Legend 1: Zone (North = red, South = blue, West = green). Legend 2: Population (M), with circles for 1, 2 and 4.

```
library(ggplot2)
```

```
cities <- data.frame(
  ID       = paste0("C", 1:8),
  Rainfall_mm = c(40, 50, 70, 70, 40, 20, 70, 50),
  Temp_C    = c(32, 40, 32, 30, 36, 26, 36, 22),
  Zone      = c("North", "South", "South", "South", "West", "South", "South", "West"),
  Population_M = c(1, 2, 4, 1, 4, 1, 1, 1)
)
```

```
cities$vj <- ifelse(cities$ID == "C4", 2.6, -1.4 - 0.4 * cities$Population_M) # label above each point (C4 below)
```

```
ggplot(cities, aes(x = Rainfall_mm, y = Temp_C, colour = Zone)) +
  geom_point(aes(size = Population_M)) +
  geom_text(aes(label = ID, vjust = vj), size = 3, colour = "black") +
  scale_size_area(max_size = 9, breaks = c(1, 2, 4), name = "Population (M)") +
  scale_colour_brewer(palette = "Set1", name = "Zone") +
  scale_x_continuous(limits = c(10, 80)) +
```

```
scale_y_continuous(limits = c(20, 42)) +
labs(x = "Rainfall (mm)", y = "Temperature (deg C)",
     title = "Rainfall vs Temperature by Zone and Population") +
theme_minimal()
```

(d) Why mapping Population_M to shape is a poor choice [2]

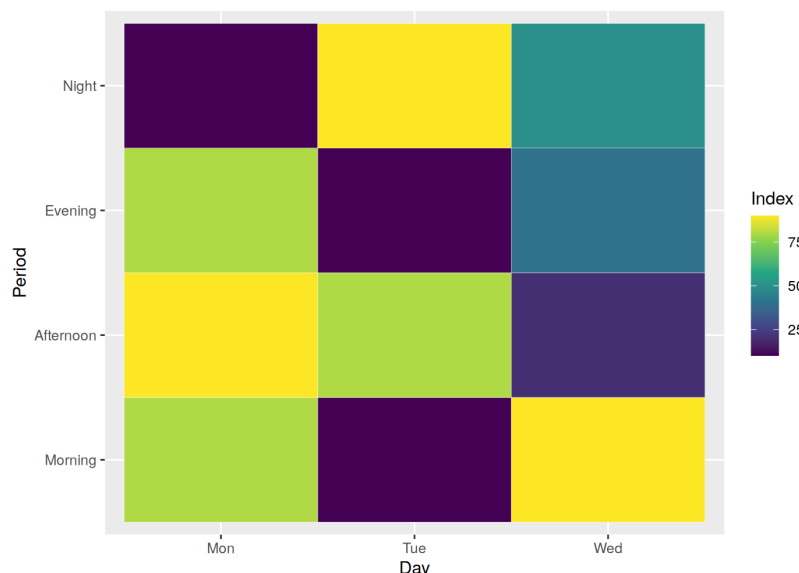
Shape is an aesthetic for **categories**: shapes have no order and no size, so the reader cannot tell that 4 is larger than 1 or by how much. Population_M is a numeric (ordered) variable, so shape would hide its magnitude; ggplot2 even refuses to map a continuous variable to shape unless it is converted to a factor. Only about six shapes can be told apart easily, and shapes of different areas can also mislead. Size (area) shows more/less directly.

Q2. R Code to Visual: Colour scale for data values [10 marks]

Reading the program. `expand.grid()` builds 12 rows (3 days x 4 periods) in which Day varies fastest, so the 12 values are filled as Mon, Tue, Wed for Morning, then for Afternoon, Evening and Night. Day and Period become factors in the order they are written, so on the y-axis **Morning is at the bottom and Night at the top**.

Period (y)	Mon	Tue	Wed
Night (top)	10 - darkest purple	90 - brightest yellow	50 - teal
Evening	80 - yellow-green	10 - darkest purple	40 - blue-green
Afternoon	90 - brightest yellow	80 - yellow-green	20 - dark blue-purple
Morning (bottom)	80 - yellow-green	10 - darkest purple	90 - brightest yellow

(a) Sketch of the chart produced [5]



Grid layout: 3 columns (Mon, Tue, Wed) and 4 rows (Morning at the bottom to Night at the top), with white lines between the tiles. Dark cells: Tue-Morning, Tue-Evening, Mon-Night (10) and Wed-Afternoon (20). Light/bright cells: the 80 and 90 values. Colour bar: titled "Index", dark purple at the bottom (about 10) to yellow at the top (about 90).

(b) Aesthetic and scale type for Day, Period and Value [3]

Variable	Aesthetic	Scale type
Day	x position	Discrete - <code>scale_x_discrete()</code>
Period	y position	Discrete - <code>scale_y_discrete()</code>
Value	fill (colour inside each tile)	Continuous colour scale - <code>scale_fill_viridis_c()</code>

The "white" in `colour = "white"` is a fixed setting for the tile borders, not a mapped variable.

(c) Is viridis sequential or diverging? One reason it suits colour-blind viewers [2]

Viridis is a **sequential** scale: it goes in one direction from dark purple (low) through teal and green to yellow (high), with no neutral middle. It suits colour-blind viewers because it changes steadily in **lightness** (dark to bright), so values can be ranked without telling red from green, and it also stays readable in greyscale.

Q3. Visual to R Code: Polar coordinates [10 marks]

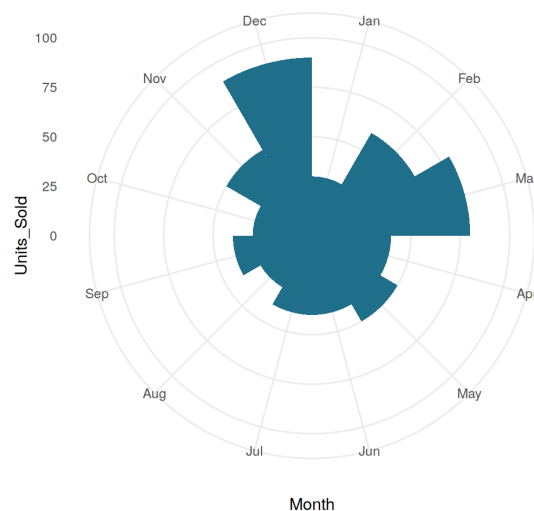
(a) Complete R program [7]

```
library(ggplot2)

sales <- data.frame(
  Month = factor(c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
    levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
    ordered = TRUE),
  Units_Sold = c(30, 60, 80, 40, 50, 40, 30, 40, 30, 50, 90)
)

ggplot(sales, aes(x = Month, y = Units_Sold)) +
  geom_col(fill = "#1F6F8B", width = 1) +
  coord_polar() +
  scale_y_continuous(limits = c(0, 100), breaks = c(0, 25, 50, 75, 100)) +
  theme_minimal()
```

How it works: (1) the data frame holds the 12 months and their Units_Sold; (2) Month is an **ordered factor** with levels Jan to Dec, so the months appear in calendar order (not alphabetical); (3) `geom_col()` draws one bar per month with its height equal to Units_Sold (width = 1 makes the bars touch); (4) `coord_polar()` turns the bars into wedges: Month becomes the angle and Units_Sold the distance from the centre; (5) `scale_y_continuous()` sets the radial axis from 0 to 100 with ticks at 0, 25, 50, 75 and 100, as in the given chart; (6) `theme_minimal()` gives the light grid.



Output of the program above (it reproduces the given chart: the largest wedges are Dec = 90 and Mar = 80, then Feb = 60).

(b) Why polar coordinates for months, and which line gives a Cartesian bar chart [3]

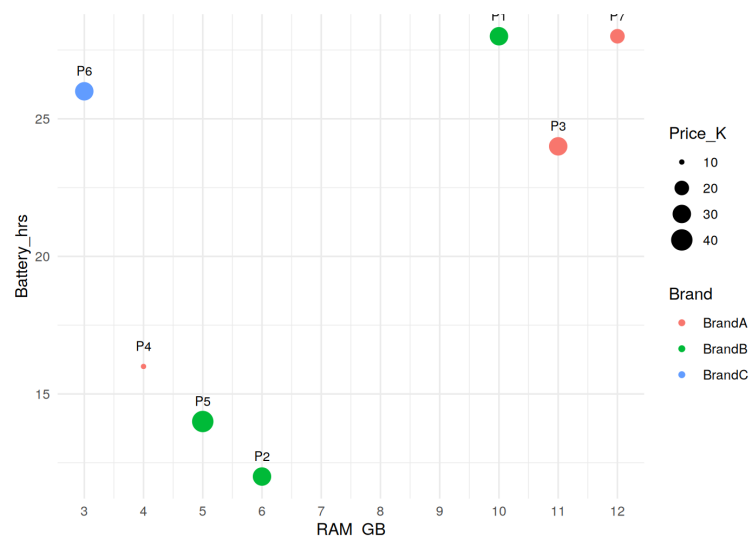
Months form a **cycle**: after December the year starts again at January. A circle shows this repeat and puts Dec next to Jan, while a straight axis places them at opposite ends. Seasonal highs and lows also appear as the shape of the wheel (here the peak around Dec to Mar and the low middle of the year). A Cartesian bar chart is obtained by **removing the line** `coord_polar()` + (or replacing it with `coord_cartesian()`): `geom_col()` then draws ordinary vertical bars with Month on the x-axis.

Q4. Visual to Table: Reading aesthetics back into data [10 marks]

(a) Reconstructed data table [5]

ID	RAM_GB	Battery_hrs	Brand	Price_K
P1	10	28	BrandB	30
P2	6	12	BrandB	30
P3	11	24	BrandA	30
P4	4	16	BrandA	10
P5	5	14	BrandB	40
P6	3	26	BrandC	30
P7	12	28	BrandA	20

How it was read: RAM_GB and Battery_hrs come from the grid lines under each point's centre; Brand from the colour (BrandA = red/salmon, BrandB = green, BrandC = blue); Price_K by matching the size of each point with the size legend: P4 is the smallest (10), P7 is a little larger (20), P1, P2, P3 and P6 are the same size as the 30 key, and P5 is the largest (40).



Check: the chart redrawn in R from this table looks the same as the given chart (same positions, colours and relative sizes).

(b) Aesthetic and scale type for each variable [3]

Variable	Aesthetic	Scale type
RAM_GB	x position	Continuous scale (ticks at 3, 4, ... 12)
Battery_hrs	y position	Continuous scale (12 to 30)
Brand	colour	Discrete (qualitative) - three colours
Price_K	size (area of the point)	Continuous - size legend 10, 20, 30, 40
ID	text label (P1 to P7)	Labels only, one per point

(c) Data type of each variable [2]

Variable	Data type
ID	Nominal (identifier / label)
RAM_GB	Discrete (whole numbers 3 to 12 GB; plotted on a continuous axis)
Battery_hrs	Continuous (hours)
Brand	Nominal
Price_K	Continuous (numeric price in thousands)

Q5. Design Decision: Choosing aesthetics and a colour palette [10 marks]

(a) Data type of each variable [2]

Variable	Data type	Reason
Aircraft_Type	Nominal	Three category names (Narrow-body / Wide-body / Regional); treated as unordered labels.
Avg_Fare	Continuous	Average fare in Rs; can take any numeric value.
Delay_Level	Ordinal	Categories with a natural order: None < Minor < Major < Severe.
Daily_Flights	Discrete	A count of flights per day (whole numbers).

(b) Aesthetic and scale type for each variable [3]

Variable	Aesthetic	Scale type
Avg_Fare	x position	Continuous - scale_x_continuous()
Daily_Flights	y position	Continuous (counts, whole-number breaks) - scale_y_continuous()
Delay_Level	colour (the required colour variable)	Discrete, ordered - scale_colour_viridis_d()
Aircraft_Type	shape	Discrete (qualitative) - scale_shape_manual()

Chart: a scatter plot with one point per route. The two numeric variables use position (the most accurate channel), the ordered variable uses colour, and the three-category variable uses shape.

```
routes$Delay_Level <- factor(routes$Delay_Level,
                             levels = c("None", "Minor", "Major", "Severe"), ordered = TRUE)

ggplot(routes, aes(x = Avg_Fare, y = Daily_Flights,
                   colour = Delay_Level, shape = Aircraft_Type)) +
  geom_point(size = 3.5) +
  scale_colour_viridis_d(direction = -1, end = 0.9) + # None = light ... Severe = dark
  scale_shape_manual(values = c(16, 17, 15)) +
  labs(x = "Average fare (Rs)", y = "Daily flights",
       colour = "Delay level", shape = "Aircraft type")
```

(c) Palette type, one R palette, and justification [3]

Palette type: **sequential**. R palette: scale_colour_viridis_d() (viridis; an alternative is scale_colour_brewer(palette = "YlGnBu")). Delay_Level is an **ordered** variable, so its colours should also be ordered: with direction = -1 the lightest colour means None and the darkest means Severe, so the reader sees at once which routes are more delayed. A diverging palette is wrong because there is no neutral middle value, and a qualitative palette would suggest the levels are unrelated and unordered. Four levels are also few enough to tell apart easily.

(d) Two steps to keep the chart readable for colour-blind viewers [2]

Step 1: use a colour-blind-safe palette that differs in lightness (viridis, as above) and avoid red-green combinations.

Step 2: do not rely on colour alone: Aircraft_Type is shown by shape as well, and large markers (size 3.5) are used so that the colours can be seen; the chart can also be checked with a colour-blindness simulator or in greyscale before it is submitted.